量子场论　第二章习题

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练习 2.1.1　（第 2.1 节，第 22 页）

设拉格朗日密度为 $\mathcal{L}(\phi_a, \partial_\mu \phi_a)$，由最小作用量原理推导场的运动方程（教材式（2.16））：

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) = 0. \tag{1}$$

回答　作用量为

$$S = \int \mathrm{d}^4 x L(\phi_a, \partial_\mu \phi_a), \tag{2}$$

作用量的变分为

$$\begin{aligned} \delta S &= \int \mathrm{d}^4 x \left(\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \phi_a\right)} \partial_\mu \left(\delta \phi_a\right) \right) \\ &= \int \mathrm{d}^4 x \left(\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \phi_a\right)} \right) \delta \phi_a + \int \mathrm{d}^4 x \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \phi_a\right)} \delta \phi_a \right), \end{aligned} \tag{3}$$

上式中散度项的四维体积分可化为边界上的三维积分，应为零。由最小作用量原理，$\delta S = 0$，再由 $\delta \phi_a$ 的任意性，得到：

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \phi_a\right)} = 0. \tag{4}$$

练习 2.2.1 （第 2.2 节，第 28 页）

查找文献阅读关于庞加莱（Poincaré）变换与相应的守恒量部分。

回答

练习 2.3.1 （第 2.3 节，第 30 页）

推导场的哈密顿正则运动方程（教材式（2.67）中的两个方程）：

$$\begin{aligned}\dot{\phi}_a &= \frac{\partial \mathcal{H}}{\partial \pi_a} - \nabla \cdot \frac{\partial \mathcal{H}}{\partial (\nabla \pi_a)}, \\ \dot{\pi}_a &= -\frac{\partial \mathcal{H}}{\partial \phi_a} + \nabla \cdot \frac{\partial \mathcal{H}}{\partial (\nabla \phi_a)}.\end{aligned} \tag{5}$$

其中 ϕ_a 为场分量，$\pi_a = \partial \mathcal{L}/\partial \dot{\phi}_a$ 为正则共轭动量，$\mathcal{H}$ 为哈密顿量密度；空间梯度项取决于 $\mathcal{H}$ 对 $\nabla\pi_a$、$\nabla\phi_a$ 的依赖。可参考教材式（2.10）的积分形式。

回答

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} = 0, \quad \pi_a = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a}, \quad \frac{\partial \mathcal{L}}{\partial \phi_a} - \dot{\pi}_a - \partial_i \frac{\partial \mathcal{L}}{\partial (\partial_i \phi_a)} = 0. \tag{6}$$

一方面，哈密顿量是拉格朗日量的勒让德变换：

$$H = \int \mathrm{d}^3 \boldsymbol{x} \pi_a \dot{\phi}_a - L, \tag{7}$$

$$\begin{aligned}\delta H &= \int \mathrm{d}^3 \boldsymbol{x} \left(\pi_a \delta \dot{\phi}_a + \dot{\phi}_a \delta \pi_a - \frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a - \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a} \delta \dot{\phi}_a - \frac{\partial \mathcal{L}}{\partial (\partial_i \phi_a)} \delta (\partial_i \phi_a) \right) \\ &= \int \mathrm{d}^3 \boldsymbol{x} \left(\dot{\phi}_a \delta \pi_a - \frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a - \frac{\partial \mathcal{L}}{\partial (\partial_i \phi_a)} \partial_i (\delta \phi_a) \right) \\ &= \int \mathrm{d}^3 \boldsymbol{x} \left(\dot{\phi}_a \delta \pi_a - \frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \partial_i \frac{\partial \mathcal{L}}{\partial (\partial_i \phi_a)} (\delta \phi_a) \right) \\ &= \int \mathrm{d}^3 \boldsymbol{x} \left(\dot{\phi}_a \delta \pi_a + \left(\partial_i \frac{\partial \mathcal{L}}{\partial (\partial_i \phi_a)} - \frac{\partial \mathcal{L}}{\partial \phi_a} \right) \delta \phi_a \right) \\ &= \int \mathrm{d}^3 \boldsymbol{x} \left(\dot{\phi}_a \delta \pi_a - \dot{\pi}_a \delta \phi_a \right),\end{aligned} \tag{8}$$

其中，分部积分后舍去了边界项。另一方面，$\mathcal{H} = \mathcal{H}(\phi_a, \pi_a, \nabla\phi_a, \nabla\pi_a)$，于是

$$\begin{aligned}\delta H &= \int \mathrm{d}^3 \boldsymbol{x} \left(\frac{\partial \mathcal{H}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{H}}{\partial \pi_a} \delta \pi_a + \frac{\partial \mathcal{H}}{\partial (\partial_i \phi_a)} \delta (\partial_i \phi_a) + \frac{\partial \mathcal{H}}{\partial (\partial_i \pi_a)} \delta (\partial_i \pi_a) \right) \\ &= \int \mathrm{d}^3 \boldsymbol{x} \left(\frac{\partial \mathcal{H}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{H}}{\partial \pi_a} \delta \pi_a + \frac{\partial \mathcal{H}}{\partial (\partial_i \phi_a)} \partial_i (\delta \phi_a) + \frac{\partial \mathcal{H}}{\partial (\partial_i \pi_a)} \partial_i (\delta \pi_a) \right) \\ &= \int \mathrm{d}^3 \boldsymbol{x} \left(\frac{\partial \mathcal{H}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{H}}{\partial \pi_a} \delta \pi_a - \partial_i \frac{\partial \mathcal{H}}{\partial (\partial_i \phi_a)} (\delta \phi_a) - \partial_i \frac{\partial \mathcal{H}}{\partial (\partial_i \pi_a)} (\delta \pi_a) \right) \\ &= \int \mathrm{d}^3 \boldsymbol{x} \left(\left(\frac{\partial \mathcal{H}}{\partial \phi_a} - \partial_i \frac{\partial \mathcal{H}}{\partial (\partial_i \phi_a)} \right) \delta \phi_a + \left(\frac{\partial \mathcal{H}}{\partial \pi_a} - \partial_i \frac{\partial \mathcal{H}}{\partial (\partial_i \pi_a)} \right) \delta \pi_a \right),\end{aligned} \tag{9}$$

其中分部积分后也舍去了边界项。二者对比可得：

$$\frac{\partial \mathcal{H}}{\partial \pi_a} - \partial_i \frac{\partial \mathcal{H}}{\partial (\partial_i \pi_a)} = \dot{\phi}_a, \quad \frac{\partial \mathcal{H}}{\partial \phi_a} - \partial_i \frac{\partial \mathcal{H}}{\partial (\partial_i \phi_a)} = -\dot{\pi}_a, \tag{10}$$

也即

$$\dot{\phi}_a = \frac{\partial \mathcal{H}}{\partial \pi_a} - \nabla \cdot \frac{\partial \mathcal{H}}{\partial (\nabla \pi_a)}, \quad \dot{\pi}_a = -\frac{\partial \mathcal{H}}{\partial \pi_a} + \nabla \cdot \frac{\partial \mathcal{H}}{\partial (\nabla \phi_a)}. \tag{11}$$